

Veterans e-Post Hub

Project Requirements and Specifications

Sponsor: Becky Buri

Whitman County Veteran Services



15-FA25-SP26-WCVS-WEB

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I. Introduction

Supporting those who have served our nation is a responsibility that extends far beyond expressions of gratitude. Many veterans encounter significant challenges as they transition to civilian life—ranging from securing reliable resources and community support to pursuing meaningful career opportunities. This project builds upon the foundation established by previous developers who shared the vision of empowering veterans through technology. Our objective is to strengthen and expand this web platform by enhancing existing functionalities and introducing new features, ensuring it remains an accessible, intuitive, and user-centered digital resource for veterans.

Our Veteran Service Officer, Becky Buri, remains deeply committed to advancing the well-being of those who have served, and this project reflects that same commitment. Serving veterans is not only central to our mission but also a guiding principle in every design decision. While the website [1] is already fully operational, our current efforts focus on elevating it to the next level—simplifying its structure, eliminating redundancies, and integrating new capabilities to deliver a streamlined and seamless experience tailored to the needs of its users.

II. System Requirements Specification

This section describes the functional and non-functional requirements of the system. Each requirement is motivated by stakeholder needs and directly supports the overall mission of providing an accessible and reliable digital platform for veterans.

II.A Stakeholders and Needs

- **Veterans (Primary Users):** Require an intuitive, accessible interface to find events, resources, and announcements relevant to their transition into civilian life.

- **Administrators (Content Managers):** Require secure, reliable tools to publish and update information (e.g., events, resources) quickly and accurately.
- **Veteran Service Officer (Client):** Requires the platform to reflect organizational values of accessibility, accountability, and reliability in serving the veteran community.
- **Developers/Maintainers (Secondary Stakeholders):** Require clear system requirements to ensure maintainability, scalability, and adherence to quality standards.

II.B. Functional Requirements

Below are the functional requirements listed by name with a description, source, and priority level.

II.B.1 Event Management

Name	FR-1: Create Events
Description	The system shall limit access to the create event tab to Becky, and as the admin, she should be the only person who can create an event.
Source	Specified by the client.
Priority	<u>Priority Level 0</u> : Essential and Required functionality

Name	FR-2: Set Date Ranges
Description	The system shall allow for date ranges to be set for created events.
Source	Specified by the client.
Priority	<u>Priority Level 0</u> : Essential and Required functionality

Name	FR-3: Set Priority Level
Description	The system shall assign priority levels to created events based on their upcoming dates.
Source	Specified by the client.

Priority	<u>Priority Level 1</u> : Desirable Functionality
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Name	FR-4: Automatic Archiving
Description	The system shall automatically archive past events, keeping them accessible in a “Past Events” section.
Source	Team recommendation after looking over the webpage.
Priority	<u>Priority Level 1</u> : Desirable Functionality

Name	FR-5: Event Flyer Downloads
Description	The system shall allow event flyers to be downloaded as PDFs for offline use.
Source	Team recommendation after looking over the webpage.
Priority	<u>Priority Level 2</u> : Stretch Feature

II.B.2 Resource Management

Name	FR-6: Resource Tab - Excel Sheet
Description	The system shall have a tab that leads to the Excel sheet Becky has created, containing all the updated contact information and correct offices.
Source	Specified by the client.
Priority	<u>Priority Level 0</u> : Essential and Required Functionality

Name	FR-7: Mann-Grandstaff Link
Description	The system shall open up the Mann-Grandstaff page when the corresponding link is clicked.
Source	Specified by the client.

Priority	<u>Priority Level 0</u> : Essential and Required Functionality
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II.B.3 User Interface

Name	FR-8: Uniform Event Tiles
Description	The system shall display tiles with uniform events on the main page.
Source	Specified by the client.
Priority	<u>Priority Level 0</u> : Essential and Required Functionality

Name	FR-9: Event Image Display
Description	The system shall display the image of a clicked-on event tile on the same webpage, without opening a new tab.
Source	Specified by the client.
Priority	<u>Priority Level 1</u> : Desirable Functionality

II.B.4 System Navigation & Search

Name	FR-10: Email Link
Description	The system shall display a new email draft to Becky when the email link is clicked, rather than redirecting to the 365 sign-in page.
Source	Team recommendation after looking over the webpage.
Priority	<u>Priority Level 0</u> : Essential and Required Functionality

Name	FR-11: Search Functionality
Description	The system shall include a search function that allows users to quickly locate resources, events, or contact information by keyword.
Source	Specified by the client.

Priority	Priority Level 1: Desirable Functionality
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II.C Non-Functional Requirements

The following are non-functional requirements outlining the qualities the website and backend must uphold. These objectives ensure that the system remains reliable, accessible, and user-friendly, even though they do not directly define its functional behavior.

- **Usability and Design:** The system shall present a clean and simple layout with properly categorized information, responsive headers and footers, consistent formatting, and readable font sizes (14–16 px for body text, 20–24 px for headings). The design shall avoid horizontal scrolling at 100 percent zoom and maintain sufficient color contrast in compliance with WCAG 2.1 AA standards.
- **Performance:** The system shall achieve page load times of less than or equal to three seconds on standard broadband connections and update newly published events, resources, and announcements within five minutes.
- **Reliability:** The system shall maintain 99% uptime on a monthly basis to ensure veterans have consistent access to time-sensitive resources.
- **Accessibility:** The system shall ensure that users, including those with disabilities, can navigate and interact with the website by adhering to accessibility standards and guidelines.
- **Accountability:** The system shall log all administrative actions (e.g., event creation) to support transparency, auditing, and oversight.

II.D Use Cases Diagram, Selected Use Cases (UC) & User Stories (US)

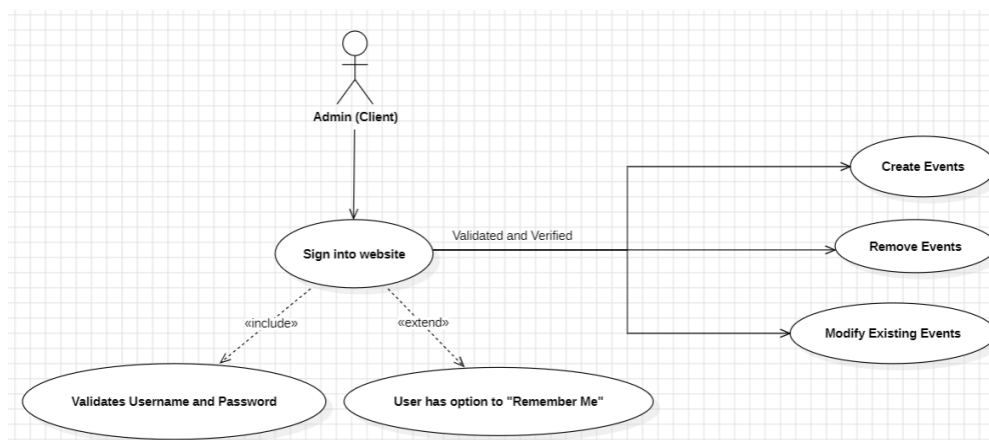


Figure 1. Use Case Diagram illustrating Admin sign-in and the Actions that the user is authorized to take.

UC-1: Admin Sign-in to Edit Events

Actors	Admin/Client
Pre-condition	The user is on the Admin sign-in page.
Post-condition	The user is successfully authorized and granted access to event management features after credential verification.
Basic Path	1. The user navigates to the Admin sign-in Page. 2. The system displays the login form with username/email and password fields. 3. The user enters valid credentials. 4. Click the "Sign In" button. 5. The system verifies the credentials against stored records. 6. Upon successful verification, the system establishes an authenticated session. 7. The system redirects the user to the Admin dashboard where event editing options are available.
Alternate Paths	3a. The user enters incorrect credentials. 4a. The user selects "Forget Password". 5a. The credentials are invalid and return to step 1.

US-01:

"As an admin, I want a streamlined event creation option so that I am not filling out unneeded information."

Feature: Admin Event Management

Scenario: User is signed in and creates an event

Given the user is authenticated as an admin

And the user is on the "Create Event" form

Create an event by filling only the required fields and relevant details

And the user submits the form

Optional fields not shown are saved as null

And the user sees a confirmation with the event tile.

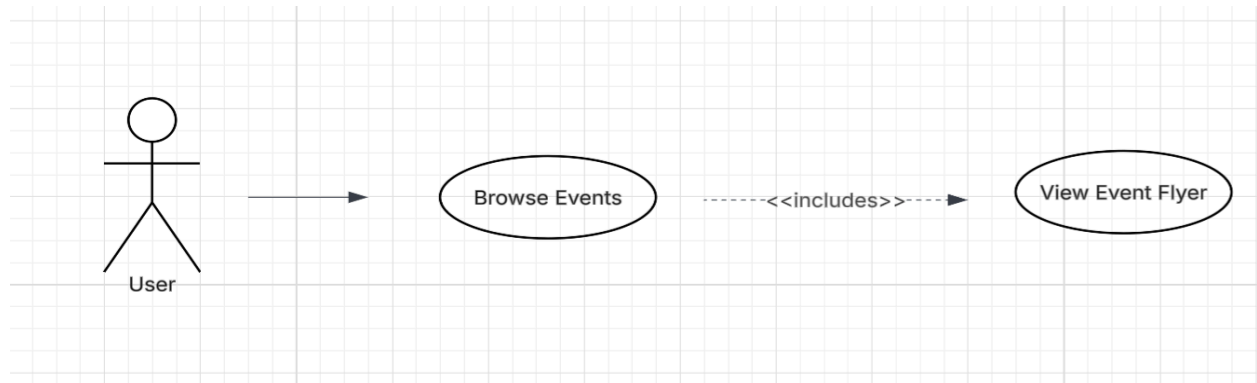


Figure 2. Use Case Diagram illustrating User interaction to view an event flyer from browsing the event tiles grid.

UC-2: View Uniform Event Flyers

Actors	General User
Pre-condition	The user is on the landing page, and the event tiles have loaded.
Post-condition	The selected event flyer is displayed on the same page, in a uniform and consistent size.
Basic Path	1. The user scrolls to the event tiles section on the main page. 2. The user clicks an event tile. 3. The system retrieves the flyer associated with the selected event. 4. The system displays the flyer on the same webpage using a fixed bounding frame to enforce uniform size. 5. The system preserves aspect ratio (fit/contain) to avoid distortion. 6. The tiles/flyer area remains uniform and readable.
Alternate Paths	2a. The system shows a placeholder ("Flyer unavailable") within the same fixed frame to preserve uniformity. 4b. The system shows a loading skeleton within the fixed dimensions until the flyer renders.

US-02:

"As a community veteran, I want the event flyers to be uniform so that they are readable."

Feature: Event flyers display uniformly.

Scenario: Event tiles load flyers in a uniform format.

Given that I have scrolled down to the event tiles on the webpage

When I click on different events, the flyers display at the same image size

Then all the event tiles appear uniform on the webpage

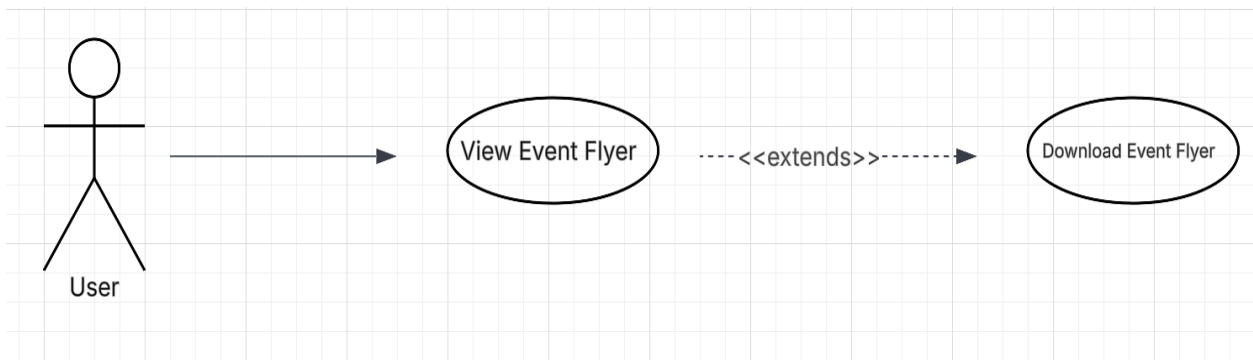


Figure 3. Use case diagram illustrating the User being able to download a PDF of the event flyer.

UC-3:Download Event Flyers

Actors	General User
Pre-condition	The user is viewing the event details, and the flyer is visible.
Post-condition	The flyer is successfully downloaded and can be opened or shared offline.
Basic Path	1. The user navigates to the event details page. 2. The system displays the event flyer on the page. 3. The user clicks the “Download Flyer (PDF)” button. 4. The system generates a PDF version of the flyer. 5. The system initiates the download of the flyer to the user’s device. 6. The user opens the file, and the PDF is readable.
Alternate Paths	2a. If the flyer is not available, the system shows a “No flyer available” message and disables the download option. 4a. The system displays an error message and prompts the user to retry. 5a. The system fails to complete the download and notifies the user to check their connection.

US-03:

“As a veteran, I want event flyers to be downloadable so that I can print or share them offline.”

Feature: Event Flyer Downloads

Scenario: Download event flyer as PDF.

Given that I am viewing an event details page
 And the event flyer is visible on the page
 When I click the “Download Flyer (PDF)” button
 Then the system generates a PDF of the flyer
 And the file begins downloading to my device
 And the downloaded file opens successfully and is readable

II.F Requirements Traceability Matrix

User Story ID	Functional Requirement(s)	Use Case(s)	Verification Method	Notes
US-01	FR-1 (Create Events), FR-2 (Set Date Ranges)	UC-1 (Admin Sign-in to Edit Events)	Functional testing, Admin login verification	Matches Becky’s admin controls
US-02	FR-8 (Uniform Event Tiles), FR-9 (Event Image Display)	UC-2 (View Uniform Event Flyers)	UI inspection, responsive test	Ensures consistent visual design
US-03	FR-5 (Event Flyer Downloads)	UC-3 (Download Event Flyers)	Download test, file validation	Confirms PDF readability
US-04	FR-4 (Automatic Archiving)	N/A	Scheduled task simulation	Reduces manual maintenance
US-05	FR-11 (Search Functionality)	N/A	Keyword query test	Enhances navigation speed
US-06	FR-6 (Resource Tab – Excel Sheet), FR-7 (Mann-Grandstaff Link)	N/A	Link and data validation	Keeps resources accurate and up-to-date
US-07	Non-Functional Requirements (Usability,	N/A	Accessibility (WCAG 2.1 AA) audit	Applies to all front-end views

	Accessibility, Reliability)			
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III. System Evolution

The development of this project is based on several fundamental assumptions regarding technology, user needs, and client expectations. While these assumptions guide the initial design, it is anticipated that both internal and external factors — such as hardware changes, evolving accessibility standards, or shifting stakeholder needs — may require adjustments in the future. Swiftly identifying these factors now helps avoid design decisions that could potentially constrain the system and future developers' ability to adapt, a concern consistent with established models of software evolution [2].

As the platform matures, it is expected that both the scale of content and the diversity of its users will increase. This growth may require refinements to the underlying database structure or migration to more scalable hosting solutions. Anticipating these potential changes now ensures that the system can evolve without major disruption, protecting both the continuity of service and the trust of its veteran user base.

In addition, the long-term success of the platform depends on its ability to remain aligned with changing client expectations and community needs. For example, the introduction of new veteran programs, shifts in government policy, or advances in web accessibility standards may require updates to both functionality and design. By embedding flexibility into the current requirements, the project creates room for adaptation, allowing future teams to extend or enhance the platform without reworking its core foundation.

IV. Glossary

Functional Requirements — Define *what* a system must accomplish by specifying the features, services, and behaviors that directly support user tasks and interactions. They describe the explicit functions the system is expected to perform.

Non-Functional Requirements — Define *how* a system should operate by establishing quality attributes, performance criteria, and constraints. They address characteristics such as usability, reliability, efficiency, and maintainability, rather than specific system behaviors.

Front-End — Refers to the client-facing portion of a software system, typically encompassing the user interface (UI) and user experience (UX). The front-end is responsible for presenting information, capturing user input, and ensuring accessibility and usability. For this project, the front-end design was developed using NextUI (now referred to as HeroUI), a React-based component library that supports responsive and accessible interface design.

Back-End — Refers to the server-side portion of a software system, responsible for implementing business logic, processing data, and managing communication with databases or external services. The back-end ensures system reliability, scalability, and security while supporting the functionality delivered through the front-end. It is typically implemented using

server-side programming languages, frameworks, and database technologies. For this project, the back-end is developed using Prisma (an open-source ORM for database management) in conjunction with SQLite queries to handle structured data storage and retrieval.

V. References

[1] "Welcome to Whitman County Veteran page," Veteran e-Post Hub, <https://whitmancountyveterans.com/> (accessed Sep. 6, 2025).

[2] "Software Engineering | Software Evolution." *GeeksforGeeks*, www.geeksforgeeks.org/software-engineering/software-engineering-software-evolution. (Accessed 27 Sept. 2025).